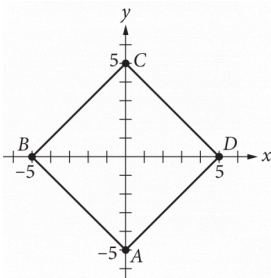


Question ID cf53cb56

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: cf53cb56



In the xy -plane shown, square $ABCD$ has its diagonals on the x - and y -axes. What is the area, in square units, of the square?

- A. 20
- B. 25
- C. 50
- D. 100

ID: cf53cb56 Answer

Correct Answer:
C

Rationale

Choice C is correct. The two diagonals of square $ABCD$ divide the square into 4 congruent right triangles, where each triangle has a vertex at the origin of the graph shown. The formula for the area of a triangle is $A = \frac{1}{2}bh$, where b is the base length of the triangle and h is the height of the triangle. Each of the 4 congruent right triangles has a height of 5 units and a base length of 5 units. Therefore, the area of each triangle is $A = \frac{1}{2}(5)(5)$, or 12.5 square units. Since the 4 right triangles are congruent, the area of each is $\frac{1}{4}$ of the area of square $ABCD$. It follows that the area of the square $ABCD$ is equal to 4×12.5 , or 50 square units.

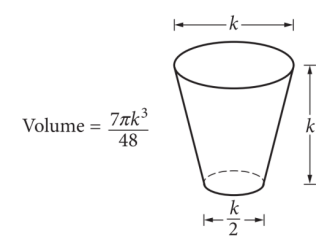
Choices A and D are incorrect and may result from using 5 or 25, respectively, as the area of one of the 4 congruent right triangles formed by diagonals of square $ABCD$. However, the area of these triangles is 12.5. Choice B is incorrect and may result from using 5 as the length of one side of square $ABCD$. However, the length of a side of square $ABCD$ is $5\sqrt{2}$.

Question Difficulty:
Medium

Question ID 37dde49f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 37dde49f



The glass pictured above can hold a maximum volume of 473 cubic centimeters, which is approximately 16 fluid ounces. What is the value of k , in centimeters?

- A. 2.52
- B. 7.67
- C. 7.79
- D. 10.11

ID: 37dde49f Answer

Correct Answer:
D

Rationale

Choice D is correct. Using the volume formula $V = \frac{7 \pi k^3}{48}$ and the given information that the volume of the glass is 473 cubic centimeters, the value of k can be found as follows:

$$473 = \frac{7 \pi k^3}{48}$$
$$k^3 = \frac{473(48)}{7 \pi}$$
$$k = \sqrt[3]{\frac{473(48)}{7 \pi}} \approx 10.10690$$

Therefore, the value of k is approximately 10.11 centimeters.

Choices A, B, and C are incorrect. Substituting the values of k from these choices in the formula results in volumes of approximately 7 cubic centimeters, 207 cubic centimeters, and 217 cubic centimeters, respectively, all of which contradict the given information that the volume of the glass is 473 cubic centimeters.

Question Difficulty:
Medium

Question ID 468613c0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 468613c0

A triangle has a base length of **56** centimeters and a height of **112** centimeters. What is the area, in square centimeters, of the triangle?

- A. **168**
- B. **1,568**
- C. **3,136**
- D. **6,272**

ID: 468613c0 Answer

Correct Answer:
C

Rationale

Choice C is correct. The area, A , of a triangle is given by the formula $A = \frac{1}{2}bh$, where b is the base length and h is the height of the triangle. It's given that a triangle has a base length of **56** centimeters and a height of **112** centimeters. Substituting **56** for b and **112** for h in the formula $A = \frac{1}{2}bh$ yields $A = (\frac{1}{2})(56)(112)$, or $A = 3,136$. Therefore, the area, in square centimeters, of the triangle is **3,136**.

Choice A is incorrect. This is the value of $56 + 112$, not $(\frac{1}{2})(56)(112)$.

Choice B is incorrect. This is the value of $(\frac{1}{4})(56)(112)$, not $(\frac{1}{2})(56)(112)$.

Choice D is incorrect. This is the value of $(56)(112)$, not $(\frac{1}{2})(56)(112)$.

Question Difficulty:
Medium

Question ID a2e76b60

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: a2e76b60

A cylindrical can containing pieces of fruit is filled to the top with syrup before being sealed. The base of the can has an area of 75 cm^2 , and the height of the can is 10 cm. If 110 cm^3 of syrup is needed to fill the can to the top, which of the following is closest to the total volume of the pieces of fruit in the can?

- A. 7.5 cm^3
- B. 185 cm^3
- C. 640 cm^3
- D. 750 cm^3

ID: a2e76b60 Answer

Correct Answer:
C

Rationale

Choice C is correct. The total volume of the cylindrical can is found by multiplying the area of the base of the can, 75 cm^2 , by the height of the can, 10 cm, which yields 750 cm^3 . If the syrup needed to fill the can has a volume of 110 cm^3 , then the remaining volume for the pieces of fruit is $750 - 110 = 640\text{ cm}^3$.

Choice A is incorrect because if the fruit had a volume of 7.5 cm^3 , there would be $750 - 7.5 = 742.5\text{ cm}^3$ of syrup needed to fill the can to the top. Choice B is incorrect because if the fruit had a volume of 185 cm^3 , there would be $750 - 185 = 565\text{ cm}^3$ of syrup needed to fill the can to the top. Choice D is incorrect because it is the total volume of the can, not just of the pieces of fruit.

Question Difficulty:
Medium

Question ID cecbdeba

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: cecbdeba

A right circular cylinder has a volume of **432** cubic centimeters. The area of the base of the cylinder is **24** square centimeters. What is the height, in centimeters, of the cylinder?

- A. 18
- B. 24
- C. 216
- D. 10,368

ID: cecbdeba Answer

Correct Answer:
A

Rationale

Choice A is correct. The volume, V , of a right circular cylinder is given by the formula $V = \pi r^2 h$, where πr^2 is the area of the base of the cylinder and h is the height. It's given that a right circular cylinder has a volume of **432** cubic centimeters and the area of the base is **24** square centimeters. Substituting **432** for V and **24** for πr^2 in the formula $V = \pi r^2 h$ yields $432 = 24h$. Dividing both sides of this equation by **24** yields $18 = h$. Therefore, the height of the cylinder, in centimeters, is **18**.

Choice B is incorrect. This is the area of the base, in square centimeters, not the height, in centimeters, of the cylinder.

Choice C is incorrect. This is the height, in centimeters, of a cylinder if its volume is **432** cubic centimeters and the area of its base is **2**, not **24**, cubic centimeters.

Choice D is incorrect. This is the height, in centimeters, of a cylinder if its volume is **432** cubic centimeters and the area of its base is $\frac{1}{24}$, not **24**, cubic centimeters.

Question Difficulty:
Medium

Question ID ae33e0a1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: ae33e0a1

A right circular cylinder has a base diameter of **22** centimeters and a height of **6** centimeters. What is the volume, in cubic centimeters, of the cylinder?

- A. **132** π
- B. **264** π
- C. **726** π
- D. **2,904** π

ID: ae33e0a1 Answer

Correct Answer:
C

Rationale

Choice C is correct. The volume, V , of a right circular cylinder is given by the formula $V = \pi r^2 h$, where r is the radius of the base of the cylinder and h is the height of the cylinder. It's given that a right circular cylinder has a height of **6** centimeters. Therefore, $h = 6$. It's also given that the cylinder has a base diameter of **22** centimeters. The radius of a circle is half the diameter of the circle. Since the base of a right circular cylinder is a circle, it follows that the radius of the base of the right circular cylinder is $\frac{22}{2}$, or **11**, centimeters. Therefore, $r = 11$. Substituting **11** for r and **6** for h in the formula $V = \pi r^2 h$ yields $V = \pi(11)^2(6)$, which is equivalent to $V = \pi(121)(6)$, or $V = 726\pi$. Therefore, the volume, in cubic centimeters, of the cylinder is **726** π .

Choice A is incorrect. This is the volume of a right circular cylinder that has a base diameter of $2\sqrt{22}$, not **22**, centimeters and a height of **6** centimeters.

Choice B is incorrect. This is the volume of a right circular cylinder that has a base diameter of $4\sqrt{11}$, not **22**, centimeters and a height of **6** centimeters.

Choice D is incorrect. This is the volume of a right circular cylinder that has a base diameter of **44**, not **22**, centimeters and a height of **6** centimeters.

Question Difficulty:
Medium

Question ID aef4fd8a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: aef4fd8a

The length of each side of a square is **94** centimeters (cm). Which expression gives the area, in **cm²**, of the square?

- A. **2 · 94**
- B. **2 · 94 · 94**
- C. **4 · 94**
- D. **94 · 94**

ID: aef4fd8a Answer

Correct Answer:

D

Rationale

Choice D is correct. The area of a square is given by **s^2** , where **s** is the length of each side of the square. It's given that the length of each side of a square is **94 cm**. It follows that the area, in **cm²**, of the square is **$(94)^2$** , or **94 · 94**. Therefore, the expression that gives the area, in **cm²**, of the square is **94 · 94**.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect. This expression gives the perimeter, in **cm**, of the square.

Question Difficulty:

Medium

Question ID 9c0a0eca

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 9c0a0eca

A triangle has a base length of **10** centimeters and a corresponding height of **70** centimeters. What is the area, in square centimeters, of the triangle?

- A. **700**
- B. **350**
- C. **175**
- D. **80**

ID: 9c0a0eca Answer

Correct Answer:
B

Rationale

Choice B is correct. The area, A , of a triangle is given by $A = (\frac{1}{2})bh$, where b is the length of a base of the triangle and h is the corresponding height of the triangle. It's given that a triangle has a base length of **10** centimeters and a corresponding height of **70** centimeters. Substituting **10** for b and **70** for h in the formula $A = (\frac{1}{2})bh$ yields $A = (\frac{1}{2})(10)(70)$, or $A = 350$. Therefore, the area, in square centimeters, of the triangle is **350**.

Choice A is incorrect. This is the product of the given base and height of the triangle, not its area.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the sum of the given base and height of the triangle, not its area.

Question Difficulty:
Medium

Question ID 08b7a3f5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 08b7a3f5

A triangular prism has a height of **8 centimeters (cm)** and a volume of **216 cm³**. What is the area, **in cm²**, of the base of the prism? (The volume of a triangular prism is equal to ***Bh***, where ***B*** is the area of the base and ***h*** is the height of the prism.)

ID: 08b7a3f5 Answer

Correct Answer:

27

Rationale

The correct answer is **27**. It's given that a triangular prism has a volume of **216 cubic centimeters (cm³)** and the volume of a triangular prism is equal to ***Bh***, where ***B*** is the area of the base and ***h*** is the height of the prism. Therefore, **216 = *Bh***. It's also given that the triangular prism has a height of **8 cm**. Therefore, ***h* = 8**. Substituting **8** for ***h*** in the equation **216 = *Bh*** yields **216 = *B*(8)**. Dividing both sides of this equation by **8** yields **27 = *B***. Therefore, the area, **in cm²**, of the base of the prism is **27**.

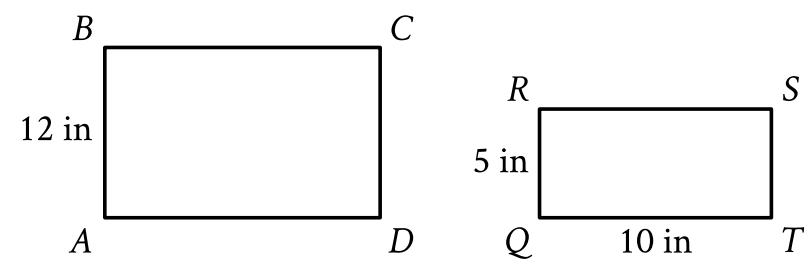
Question Difficulty:

Medium

Question ID e9c5bfb2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: e9c5bfb2



Note: Figure not drawn to scale.

Rectangles $ABCD$ and $QRST$ shown are similar, where A, B, C , and D correspond to Q, R, S , and T , respectively. What is the length, in inches (**in**), of $\overline{AD}$?

- A. 60
- B. 24
- C. 17
- D. 10

ID: e9c5bfb2 Answer

Correct Answer:

B

Rationale

Choice B is correct. It's given that rectangles $ABCD$ and $QRST$ are similar, where A, B, C , and D correspond to Q, R, S , and T , respectively. It follows that $\overline{AB}$ corresponds to $\overline{QR}$ and $\overline{AD}$ corresponds to $\overline{QT}$. If two rectangles are similar, then the lengths of their corresponding sides are proportional. It's given in the figure that the length of $\overline{AB}$ is **12** inches, the length of $\overline{QR}$ is **5** inches, and the length of $\overline{QT}$ is **10** inches. If x is the length, in inches, of $\overline{AD}$, then $\frac{12}{5}$ is equivalent to $\frac{x}{10}$. Therefore, the value of x can be found using the equation $\frac{12}{5} = \frac{x}{10}$. Multiplying each side of this equation by **10** yields $\frac{120}{5} = x$, or **24** = x . Therefore, the length, in inches, of $\overline{AD}$ is **24**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the length, in inches, of $\overline{QT}$, not $\overline{AD}$.

Question Difficulty:

Medium

Question ID 6ae1360d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 6ae1360d

A circle has a radius of **2.1** inches. The area of the circle is $b\pi$ square inches, where b is a constant. What is the value of b ?

ID: 6ae1360d Answer

Correct Answer:
4.41, 441/100

Rationale

The correct answer is **4.41**. The area, A , of a circle is given by the formula $A = \pi r^2$, where r is the radius of the circle. It's given that the area of the circle is $b\pi$ square inches, where b is a constant, and the radius of the circle is **2.1** inches. Substituting $b\pi$ for A and **2.1** for r in the formula $A = \pi r^2$ yields $b\pi = \pi(2.1^2)$. Dividing both sides of this equation by π yields $b = 4.41$. Therefore, the value of b is **4.41**.

Question Difficulty:
Medium

Question ID 98c12e38

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 98c12e38

Circle K has a radius of 4 millimeters (mm). Circle L has an area of $100\pi \text{ mm}^2$. What is the total area, in mm^2 , of circles K and L ?

- A. 14π
- B. 28π
- C. 56π
- D. 116π

ID: 98c12e38 Answer

Correct Answer:
D

Rationale

Choice D is correct. The area, A , of a circle is given by the formula $A = \pi r^2$, where r represents the radius of the circle. It's given that circle K has a radius of 4 millimeters (mm). Substituting 4 for r in the formula $A = \pi r^2$ yields $A = \pi(4)^2$, or $A = 16\pi$. Therefore, the area of circle K is $16\pi \text{ mm}^2$. It's given that circle L has an area of $100\pi \text{ mm}^2$. Therefore, the total area, in mm^2 , of circles K and L is $16\pi + 100\pi$, or 116π .

Choice A is incorrect. This is the sum of the radii, in mm, of circles K and L multiplied by π , not the total area, in mm^2 , of the circles.

Choice B is incorrect. This is the sum of the diameters, in mm, of circles K and L multiplied by π , not the total area, in mm^2 , of the circles.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Medium

Question ID 38517165

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 38517165

A circle has a circumference of 31π centimeters. What is the diameter, in centimeters, of the circle?

ID: 38517165 Answer

Correct Answer:

31

Rationale

The correct answer is **31**. The circumference of a circle is equal to $2\pi r$ centimeters, where r represents the radius, in centimeters, of the circle, and the diameter of the circle is equal to $2r$ centimeters. It's given that a circle has a circumference of 31π centimeters. Therefore, $31\pi = 2\pi r$. Dividing both sides of this equation by π yields $31 = 2r$. Since the diameter of the circle is equal to $2r$ centimeters, it follows that the diameter, in centimeters, of the circle is **31**.

Question Difficulty:

Medium

Question ID e0874bc2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: e0874bc2

The table gives the perimeters of similar triangles TUV and XYZ , where $\overline{TU}$ corresponds to $\overline{XY}$. The length of $\overline{TU}$ is 18.

	Perimeter
Triangle TUV	37
Triangle XYZ	333

What is the length of $\overline{XY}$?

- A. 2
- B. 18
- C. 55
- D. 162

ID: e0874bc2 Answer

Correct Answer:
D

Rationale

Choice D is correct. It's given that triangle XYZ is similar to triangle TUV . Therefore, each side of triangle XYZ is k times its corresponding side of triangle TUV , where k is a constant. It follows that the perimeter of triangle XYZ is k times the perimeter of triangle TUV . It's also given that $\overline{TU}$ corresponds to $\overline{XY}$ and the length of $\overline{TU}$ is 18. Let x represent the length of $\overline{XY}$. It follows that $x = 18k$. The table shows that the perimeters of triangles TUV and XYZ are 37 and 333, respectively. It follows that $333 = 37k$, or $9 = k$. Substituting 9 for k in the equation $x = 18k$ yields $x = (18)(9)$, or $x = 162$. Therefore, the length of $\overline{XY}$ is 162.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the length of $\overline{TU}$, not the length of $\overline{XY}$.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Medium

Question ID a2659088

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: a2659088

A right circular cylinder has a height of **8 meters (m)** and a base with a radius of **12 m**. What is the volume, **in m³**, of the cylinder?

- A. 8π
- B. 20π
- C. 768π
- D. $1,152\pi$

ID: a2659088 Answer

Correct Answer:
D

Rationale

Choice D is correct. The volume, V , of a right circular cylinder is given by $V = \pi r^2 h$, where r is the radius of the circular base and h is the height of the cylinder. It's given that the cylinder has a height of **8 meters** and a base with a radius of **12 meters**. Substituting **12** for r and **8** for h in $V = \pi r^2 h$ yields $V = \pi(12)^2(8)$, or $V = 1,152\pi$. Therefore, the volume, in **m³**, of the cylinder is **1,152π**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the volume, in **m³**, of a cylinder with a radius of **8 meters** and a height of **12 meters**.

Question Difficulty:
Medium

Question ID d621cffb

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: d621cffb

A sphere has a radius of $\frac{17}{5}$ feet. What is the volume, in cubic feet, of the sphere?

- A. $\frac{5\pi}{17}$
- B. $\frac{68\pi}{15}$
- C. $\frac{32\pi}{5}$
- D. $\frac{19,652\pi}{375}$

ID: d621cffb Answer

Correct Answer:

D

Rationale

Choice D is correct. The volume, V , of a sphere can be found using the formula $V = \frac{4}{3}\pi r^3$, where r is the radius of the sphere. It's given that the sphere has a radius of $\frac{17}{5}$ feet. Substituting $\frac{17}{5}$ for r in the formula $V = \frac{4}{3}\pi r^3$ yields $V = \frac{4}{3}\pi\left(\frac{17}{5}\right)^3$, which is equivalent to $V = \frac{4}{3}\pi\left(\frac{4,913}{125}\right)$, or $V = \frac{19,652\pi}{375}$. Therefore, the volume, in cubic feet, of the sphere is $\frac{19,652\pi}{375}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the volume, in cubic feet, of a sphere with a radius of $\sqrt[3]{\frac{17}{5}}$ feet.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Medium

Question ID 151eda3c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 151eda3c

A manufacturing company produces two sizes of cylindrical containers that each have a height of 50 centimeters. The radius of container A is 16 centimeters, and the radius of container B is 25% longer than the radius of container A. What is the volume, in cubic centimeters, of container B?

- A. $16,000 \pi$
- B. $20,000 \pi$
- C. $25,000 \pi$
- D. $31,250 \pi$

ID: 151eda3c Answer

Correct Answer:
B

Rationale

Choice B is correct. If the radius of container A is 16 centimeters and the radius of container B is 25% longer than the radius of container A, then the radius of container B is $16 + (0.25)(16) = 20$ centimeters. The volume of a cylinder is $\pi r^2 h$, where r is the radius of the cylinder and h is its height. Substituting $r = 20$ and $h = 50$ into $\pi r^2 h$ yields that the volume of cylinder B is $\pi(20)^2(50) = 20,000 \pi$ cubic centimeters.

Choice A is incorrect and may result from multiplying the radius of cylinder B by the radius of cylinder A rather than squaring the radius of cylinder B. Choice C is incorrect and may result from multiplying the radius of cylinder B by 25 rather than squaring it. Choice D is incorrect and may result from taking the radius of cylinder B to be 25 centimeters rather than 20 centimeters.

Question Difficulty:
Medium

Question ID 03c6994f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 03c6994f

Square A has side lengths that are **246** times the side lengths of square B. The area of square A is ***k*** times the area of square B. What is the value of ***k***?

- A. **60,516**
- B. **492**
- C. **246**
- D. **123**

ID: 03c6994f Answer

Correct Answer:
A

Rationale

Choice A is correct. The area of a square is s^2 , where s is the side length of the square. Therefore, the area of square B is b^2 , where b is the side length of square B. It's given that square A has side lengths that are **246** times the side lengths of square B. Therefore, the side length of square A can be represented by the expression **246*b***. It follows that the area of square A is **$(246b)^2$** , or **$60,516b^2$** . It's given that the area of square A is ***k*** times the area of square B, so **$60,516b^2 = kb^2$** . Therefore, the value of ***k*** is **60,516**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Medium

Question ID c0586eb5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: c0586eb5

A cylinder has a diameter of **8** inches and a height of **12** inches. What is the volume, in cubic inches, of the cylinder?

- A. 16π
- B. 96π
- C. 192π
- D. 768π

ID: c0586eb5 Answer

Correct Answer:

C

Rationale

Choice C is correct. The base of a cylinder is a circle with a diameter equal to the diameter of the cylinder. The volume, V , of a cylinder can be found by multiplying the area of the circular base, A , by the height of the cylinder, h , or $V = Ah$. The area of a circle can be found using the formula $A = \pi r^2$, where r is the radius of the circle. It's given that the diameter of the cylinder is **8** inches. Thus, the radius of this circle is **4** inches. Therefore, the area of the circular base of the cylinder is $A = \pi(4)^2$, or 16π square inches. It's given that the height h of the cylinder is **12** inches. Substituting 16π for A and **12** for h in the formula $V = Ah$ gives $V = 16\pi(12)$, or 192π cubic inches.

Choice A is incorrect. This is the area of the circular base of the cylinder.

Choice B is incorrect and may result from using **8**, instead of **16**, as the value of r^2 in the formula for the area of a circle.

Choice D is incorrect and may result from using **8**, instead of **4**, for the radius of the circular base.

Question Difficulty:

Medium

Question ID e4b4e9ea

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: e4b4e9ea

The length of the edge of the base of a right square prism is **6** units. The volume of the prism is **2,880** cubic units. What is the height, in units, of the prism?

- A. $4\sqrt{30}$
- B. **36**
- C. $24\sqrt{5}$
- D. **80**

ID: e4b4e9ea Answer

Correct Answer:
D

Rationale

Choice D is correct. The volume, V , of a right square prism is given by the formula $V = s^2h$, where s represents the length of the edge of the base and h represents the height of the prism. It's given that the volume of a right square prism is **2,880** cubic units and the length of the edge of the base is **6** units. Substituting **2,880** for V and **6** for s in the formula $V = s^2h$ yields $2,880 = (6^2)h$, or $2,880 = 36h$. Dividing both sides of this equation by **36** yields $80 = h$. Therefore, the height, in units, of the prism is **80**.

Choice A is incorrect. This is the height, in units, of a right square prism where the length of the edge of the base is **6** units and the volume of the prism is $144\sqrt{30}$, not **2,880**, units.

Choice B is incorrect. This is the area, in square units, of the base, not the height, in units, of the prism.

Choice C is incorrect. This is the height, in units, of a right square prism where the length of the edge of the base is **6** units and the volume of the prism is $864\sqrt{5}$, not **2,880**, units.

Question Difficulty:
Medium

Question ID e336a1d2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: e336a1d2

A cube has an edge length of **41** inches. What is the volume, in cubic inches, of the cube?

- A. **164**
- B. **1,681**
- C. **10,086**
- D. **68,921**

ID: e336a1d2 Answer

Correct Answer:
D

Rationale

Choice D is correct. The volume, V , of a cube can be found using the formula $V = s^3$, where s is the edge length of the cube. It's given that a cube has an edge length of **41** inches. Substituting **41** inches for s in this equation yields $V = 41^3$ cubic inches, or $V = 68,921$ cubic inches. Therefore, the volume of the cube is **68,921** cubic inches.

Choice A is incorrect. This is the perimeter, in inches, of the cube.

Choice B is incorrect. This is the area, in square inches, of a face of the cube.

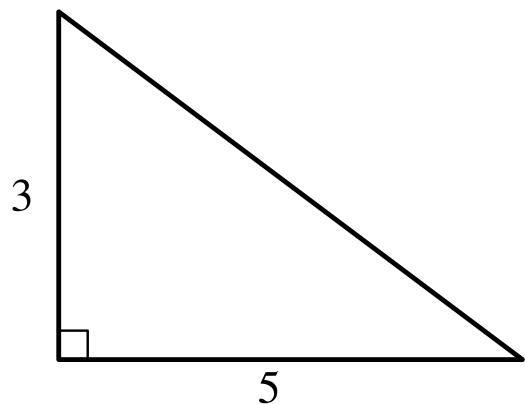
Choice C is incorrect. This is the surface area, in square inches, of the cube.

Question Difficulty:
Medium

Question ID a4ed5285

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: a4ed5285



Note: Figure not drawn to scale.

The figure shows the lengths, in inches, of two sides of a right triangle. What is the area of the triangle, in square inches?

ID: a4ed5285 Answer

Correct Answer:

7.5, 15/2

Rationale

The correct answer is $\frac{15}{2}$. The area, A , of a triangle is given by the formula $A = \frac{1}{2}bh$, where b is the length of the base of the triangle and h is the height of the triangle. In the right triangle shown, the length of the base of the triangle is 5 inches, and the height is 3 inches. It follows that $b = 5$ and $h = 3$. Substituting 5 for b and 3 for h in the formula $A = \frac{1}{2}bh$ yields $A = \frac{1}{2}(5)(3)$, which is equivalent to $A = \frac{1}{2}(15)$, or $A = \frac{15}{2}$. Therefore, the area of the triangle, in square inches, is $\frac{15}{2}$. Note that 15/2 and 7.5 are examples of ways to enter a correct answer.

Question Difficulty:

Medium

Question ID 3b931fb0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 3b931fb0

A right circular cylinder has a volume of **377** cubic centimeters. The area of the base of the cylinder is **13** square centimeters. What is the height, in centimeters, of the cylinder?

ID: 3b931fb0 Answer

Correct Answer:
29

Rationale

The correct answer is **29**. The volume, V , of a right circular cylinder is given by the formula $V = \pi r^2 h$, where r is the radius of the base of the cylinder and h is the height of the cylinder. Since the base of the cylinder is a circle with radius r , the area of the base of the cylinder is πr^2 . It's given that a right circular cylinder has a volume of **377** cubic centimeters; therefore, $V = 377$. It's also given that the area of the base of the cylinder is **13** square centimeters; therefore, $\pi r^2 = 13$. Substituting **377** for V and **13** for πr^2 in the formula $V = \pi r^2 h$ yields $377 = 13h$. Dividing both sides of this equation by **13** yields $29 = h$. Therefore, the height of the cylinder, in centimeters, is **29**.

Question Difficulty:
Medium

Question ID ec5d4823

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: ec5d4823

What is the volume, in cubic centimeters, of a right rectangular prism that has a length of 4 centimeters, a width of 9 centimeters, and a height of 10 centimeters?

ID: ec5d4823 Answer

Rationale

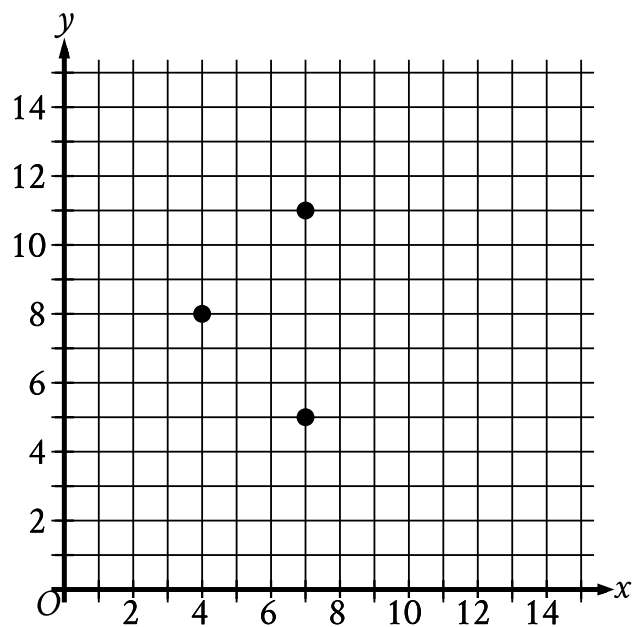
The correct answer is 360. The volume of a right rectangular prism is calculated by multiplying its dimensions: length, width, and height. Multiplying the values given for these dimensions yields a volume of $(4)(9)(10) = 360$ cubic centimeters.

Question Difficulty:
Medium

Question ID 096c7ef5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 096c7ef5



- The three points shown define a circle. The circumference of this circle is $k\pi$, where k is a constant. What is the value of k ?
- A. 3
 - B. 6
 - C. 7
 - D. 9

ID: 096c7ef5 Answer

Correct Answer:
B

Rationale

Choice B is correct. It's given that the three points shown define a circle, so the center of that circle is an equal distance from each of the three points. The point $(7, 8)$ is halfway between the points $(7, 5)$ and $(7, 11)$ and is a distance of **3** units from each of those two points. The point $(7, 8)$ is also a distance of **3** units from $(4, 8)$. Because the point $(7, 8)$ is the same distance from all three given points, it must be the center of the circle. The radius of a circle is the distance from the center to any point on the circle. Since that distance is **3**, it follows that the radius of the circle is **3**. The circumference of a circle with radius r is equal to $2\pi r$. It follows that the circumference of the circle is $2(\pi)(3)$, or 6π . It's given that the circumference of the circle is $k\pi$. Therefore, the value of k is **6**.

Choice A is incorrect. This is the radius of the circle, not the value of k in the expression $k\pi$.

Choice C is incorrect. This is the x-coordinate of the center of the circle, not the value of k in the expression $k\pi$.

Choice D is incorrect. This is the value of k for which $k\pi$ represents the area of the circle, in square units, not the circumference of the circle, in units.

Question Difficulty:

Medium

Question ID e86f0651

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: e86f0651

A circle has a radius of **43** meters. What is the area, in square meters, of the circle?

- A. $\frac{43\pi}{2}$
- B. 43π
- C. 86π
- D. $1,849\pi$

ID: e86f0651 Answer

Correct Answer:

D

Rationale

Choice D is correct. The area, **A**, of a circle is given by the formula $A = \pi r^2$, where **r** is the radius of the circle. It's given that the circle has a radius of **43** meters. Substituting **43** for **r** in the formula $A = \pi r^2$ yields $A = \pi(43)^2$, or $A = 1,849\pi$. Therefore, the area, in square meters, of the circle is **1,849π**.

Choice A is incorrect. This is the area, in square meters, of a circle with a radius of $\sqrt{\frac{43}{2}}$ meters.

Choice B is incorrect. This is the area, in square meters, of a circle with a radius of $\sqrt{43}$ meters.

Choice C is incorrect. This is the circumference, in meters, of the circle.

Question Difficulty:

Medium

Question ID 1f0b582e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 1f0b582e

Square X has a side length of **12** centimeters. The perimeter of square Y is **2** times the perimeter of square X. What is the length, in centimeters, of one side of square Y?

- A. **6**
- B. **10**
- C. **14**
- D. **24**

ID: 1f0b582e Answer

Correct Answer:
D

Rationale

Choice D is correct. The perimeter, P , of a square can be found using the formula $P = 4s$, where s is the length of each side of the square. It's given that square X has a side length of **12** centimeters. Substituting **12** for s in the formula for the perimeter of a square yields $P = 4(12)$, or $P = 48$. Therefore, the perimeter of square X is **48** centimeters. It's also given that the perimeter of square Y is **2** times the perimeter of square X. Therefore, the perimeter of square Y is **2(48)**, or **96**, centimeters. Substituting **96** for P in the formula $P = 4s$ gives $96 = 4s$. Dividing both sides of this equation by **4** gives $24 = s$. Therefore, the length of one side of square Y is **24** centimeters.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Medium

Question ID 5afbdc8e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 5afbdc8e

What is the length of one side of a square that has the same area as a circle with radius 2 ?

- A. 2
- B. $\sqrt{2\pi}$
- C. $2\sqrt{\pi}$
- D. 2π

ID: 5afbdc8e Answer

Correct Answer:
C

Rationale

Choice C is correct. The area A of a circle with radius r is given by the formula $A = \pi r^2$. Thus, a circle with radius 2 has area $\pi(2^2)$, which can be rewritten as 4π . The area of a square with side length s is given by the formula $A = s^2$. Thus, if a square has the same area as a circle with radius 2, then $s^2 = 4\pi$. Since the side length of a square must be a positive number, taking the square root of both sides of $s^2 = 4\pi$ gives $s = \sqrt{4\pi}$. Using the properties of square roots, $\sqrt{4\pi}$ can be rewritten as $(\sqrt{4})(\sqrt{\pi})$, which is equivalent to $2\sqrt{\pi}$. Therefore, $s = 2\sqrt{\pi}$.

Choice A is incorrect. The side length of the square isn't equal to the radius of the circle. Choices B and D are incorrect and may result from incorrectly simplifying the expression $\sqrt{4\pi}$.

Question Difficulty:
Medium

Question ID f67e4efc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: f67e4efc

A right circular cylinder has a volume of 45π . If the height of the cylinder is 5, what is the radius of the cylinder?

- A. 3
- B. 4.5
- C. 9
- D. 40

ID: f67e4efc Answer

Correct Answer:

A

Rationale

Choice A is correct. The volume of a right circular cylinder with a radius of r is the product of the area of the base, πr^2 , and the height, h . The volume of the right circular cylinder described is 45π and its height is 5. If the radius is r , it follows that $45\pi = \pi(r)^2(5)$. Dividing both sides of this equation by 5π yields $9 = r^2$. Taking the square root of both sides yields $r = 3$ or $r = -3$. Since r represents the radius, the value must be positive. Therefore, the radius is 3.

Choice B is incorrect and may result from finding that the square of the radius is 9, but then from dividing 9 by 2, rather than taking the square root of 9. Choice C is incorrect. This represents the square of the radius. Choice D is incorrect and may result from solving the equation $45\pi = \pi(r)^2(5)$ for r^2 , not r , by dividing by π on both sides and then by subtracting, not dividing, 5 from both sides.

Question Difficulty:

Medium

Question ID 4b7bb316

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 4b7bb316

The length of each edge of a box is **29** inches. Each side of the box is in the shape of a square. The box does not have a lid. What is the exterior surface area, in square inches, of this box without a lid?

ID: 4b7bb316 Answer

Correct Answer:
4205

Rationale

The correct answer is **4,205**. The exterior surface area of a figure is the sum of the areas of all its faces. It's given that the box does not have a lid and that each side of the box is in the shape of a square. Therefore, the box consists of **5** congruent square faces. It's also given that the length of each edge is **29** inches. Let ***s*** represent the length of an edge of a square. It follows that the area of a square is equal to ***s*²**. Therefore, the area of each of the **5** square faces is equal to **29²**, or **841**, square inches. Since the box consists of **5** congruent square faces, it follows that the sum of the areas of all its faces, or the exterior surface area of this box without a lid, is **5(841)**, or **4,205**, square inches.

Question Difficulty:
Medium